When Answers Get Cheap

– *AI, Work, and the Next Industrial Transformation to Ecosystem*

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From Physics to Industry and Ecosystem Governance

Founder, QERA (research and practice platform spanning strategy, technology and governance)
Research Fellow, Value Research Center, Doshisha University
(From Next year) Visiting Fellow, TU Delft Univ,

Quantum Physics (Post Doc)

Simulated a new quantum phenomena with super-computer

Industry Transformation / Business Planning

More than fifteen years of technology strategy, system and organisational design, and large-scale transformation across automotive, manufacturing, construction, logistics, and retail.

<Ecosystem governance>

Current research focuses on ecosystem governance and the governability of multi-agent, human–AI, and physical-AI systems, particularly where authority, responsibility, knowledge, and control are distributed across multiple actors.

1

How Does AI Change What We Value?

What Is AI's Fundamental Value?
What Does AI Actually Change?

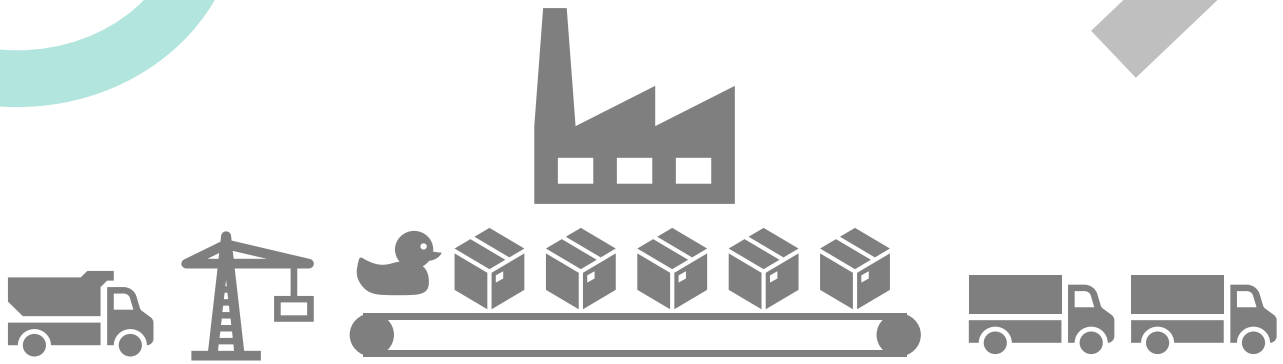
- ✓ More efficiency?
- ✓ Jobs disappearing?
- ✓ People no longer thinking?

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The Industrial Revolution: An Age of Standardization and Uniformity

Mass production of identical goods
→ efficiency

Diversity
→ cost and inefficiency



A society governed by static rules

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What People Said When IT Was Introduced

(Excel and accounting systems)

If people stop calculating by hand, they won't understand the numbers, spot errors or develop true expertise!

Sometimes the row and column totals do not match! It must be wrong!

Even with search, how can we know whether information is correct?

If it always produces the same answer, people will become boring too!

(Digital music and CDs)

It's not real sound! Children shouldn't listen to it!

Digital makes copying so easy—everyone will forget things! Not safety!

Programming? Following commands? A world of zeros and ones? Reality is full of gray areas! What happened to human judgment and thinking?

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Future Society — How Will Thinking and Values Change?

A society in which AI is taken for granted may change many ways of thinking and creating value. Here are some examples.

Shifts in Thinking and Value

Conventional society		AI-native society
Data as a record	→	Data as a real-time asset
Work = tasks	→	Work = decisions, planning, creativity and expression
Average / mass market	→	Individualization
Categorization	→	Individualized, continuous and nuanced
Memory and structured thinking	→	Framing questions
Passive consumption	→	Co-creation
Linear careers	→	Diversified, portfolio careers

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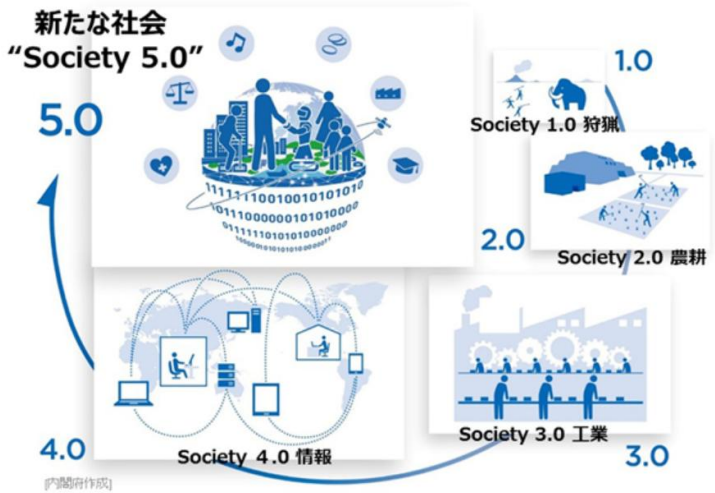
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What Changes in society?

Future Society — One Vision: Society 5.0

A future is being envisioned in which the physical world and technology converge.





Illustrative source diagram (Japanese labels)

Society 5.0 — Japanese government concept
Diagram: Ministry of Education, Culture, Sports, Science and Technology
Source: Science Portal, white paper on science and technology innovation

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Future Society — Humans Coexisting with Robots & AI

Sustainable societies are exploring new ways for technology, people and nature to coexist.

Facing global population growth, resource scarcity (water, forests, energy and food), environmental damage and climate change:

- Redesign of mobility and cities

Fewer parking areas and traffic lights

- Smart cities

Optimized energy and transportation

- Humans collaborating with robots

  People working alongside many forms of robots

- Personalization

  Healthcare, education and more



Generated with ChatGPT

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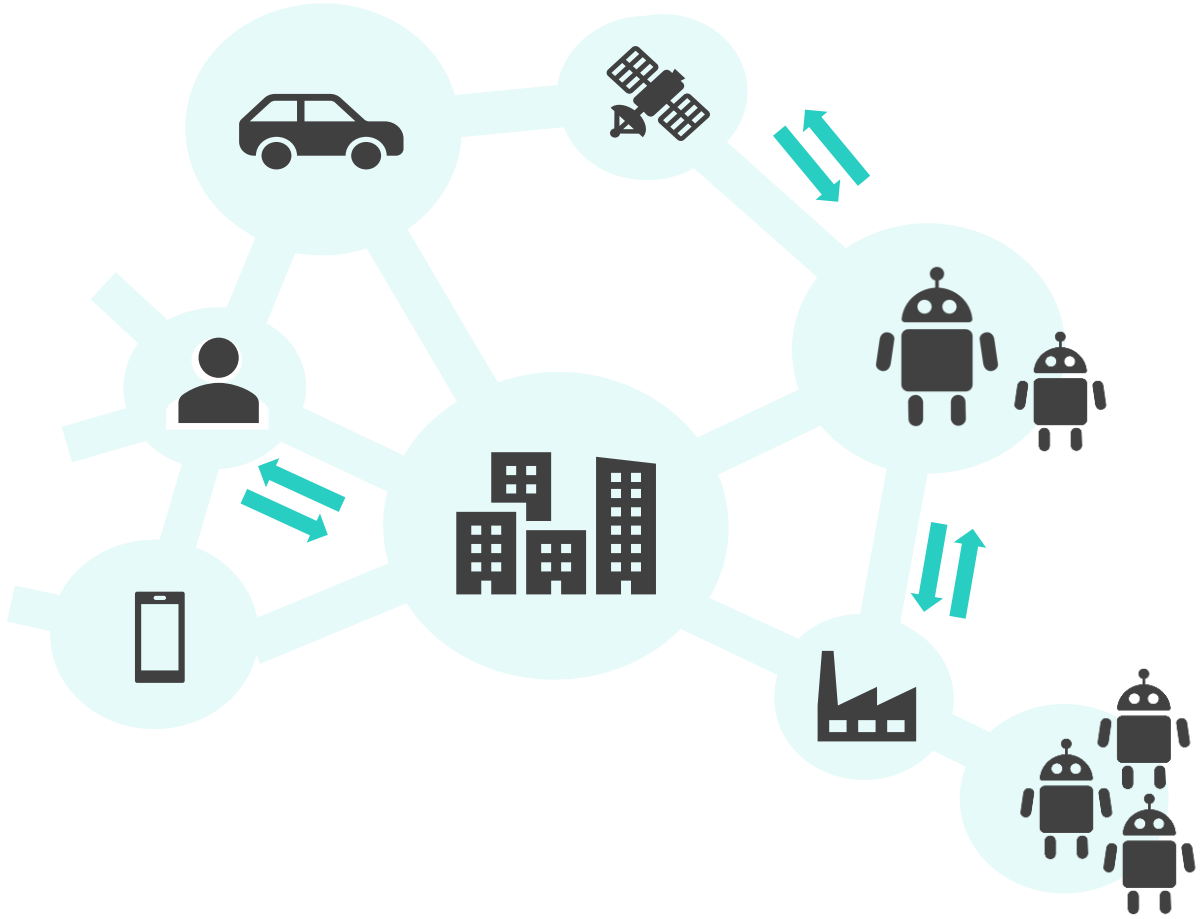
What changes in society
- Ecosystem

Business and society already operate as ecosystems

An ecosystem is not simply “many organisations”. It is a system of interdependence.

An outcome can depend on:

- ✓ multiple AI / technical components
- ✓ multiple organisations and contracts
- ✓ operating processes and human decisions
- ✓ incentives and decision rights
- ✓ rules, standards and institutions

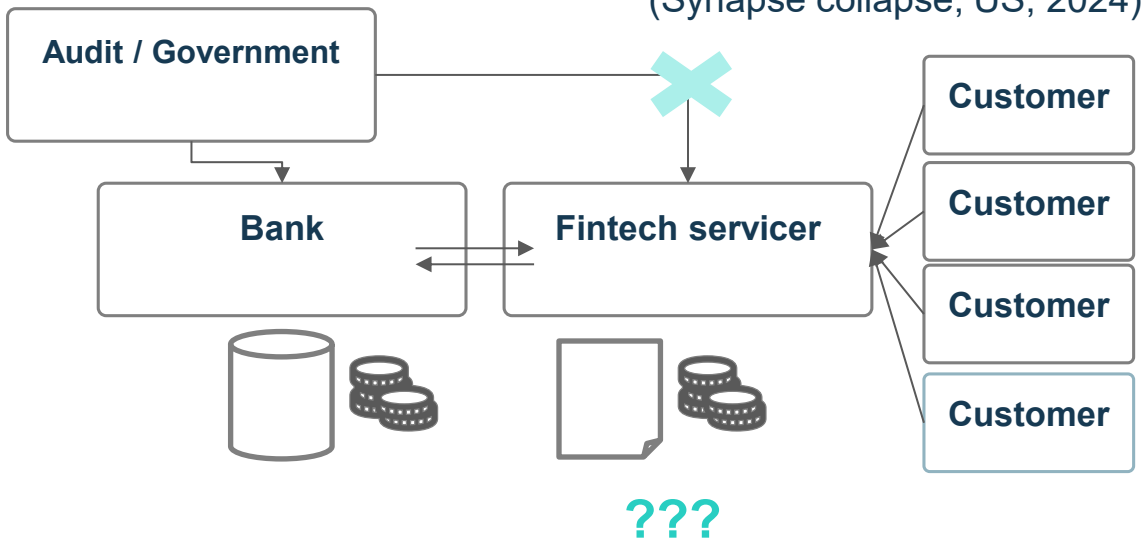


When Governance Controls Lose Their Effectiveness

In Non-direct relationship, it might be difficult to keep the condition governable

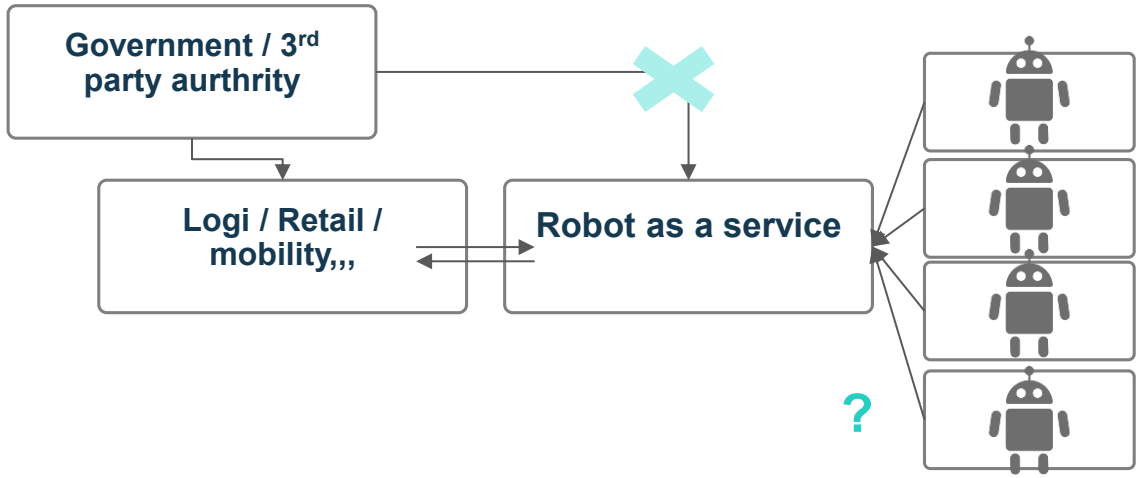
Example: Fintech Service (Illustrative)

(Synapse collapse, US, 2024)



No single actor maintained a reliable view of customer funds across the ecosystem.

Could the Same Happen in Physical AI?



Safety???

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What interacting autonomous systems already reveal

Observed incidents show shared infrastructure dependency, cascading effects and accountability gaps

WAYMO · SF blackout · 2025

A fire at a power station cut electricity to about one-third of San Francisco, so the traffic lights went dark. Waymo's self-driving cars were built to treat a dark light like a stop sign. That caused the heavy traffic jams.

Waymo, Dec 23 2025

CRUISE · North Beach · 2023

There was a big music festival nearby, and about 10 Cruise robot taxis suddenly stopped on a narrow street and blocked the traffic. Cruise said the reason was that the huge crowd was using so many phones that the mobile network got too busy, and the cars could not talk to their remote helpers quickly enough.

SF Standard, Aug 13 / KTVU, Aug 16

Amazon· 2011

A book about flies was priced at about 23 million dollars on Amazon. Two sellers used computer programs to set their prices by looking at each other's price. One always went a little lower, the other always went much higher, so the price kept going up every day.

Amazon 2011

WAYMO · honking · 2024

Waymo cars waited in a parking lot in San Francisco at night. Each car had a feature that honks its horn when another car comes too close, to avoid an accident. Because many cars were moving close together, they kept honking at each other all night and woke up the neighbours, so Waymo changed the software.

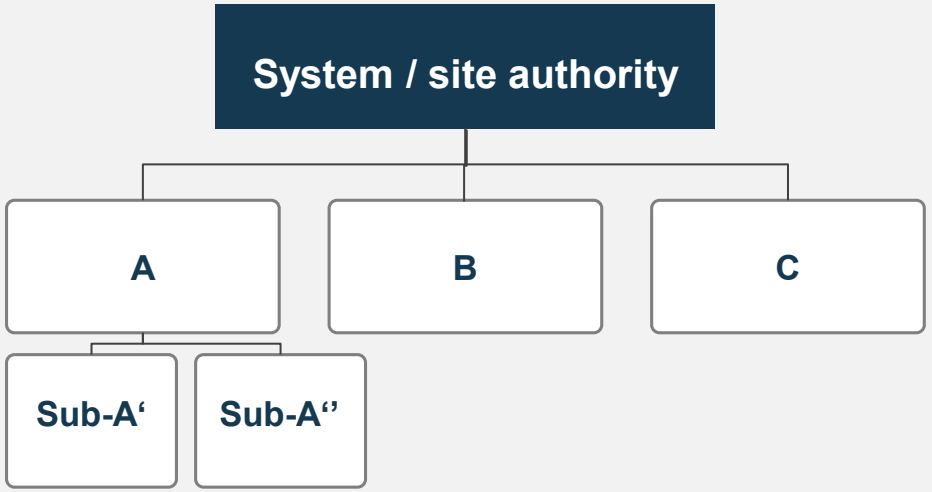
Axios, Aug 13 2024

These illustrate the need for a collective response across society and the industry

01 Distributed / Fragmented Authority

Who can decide / require / verify?

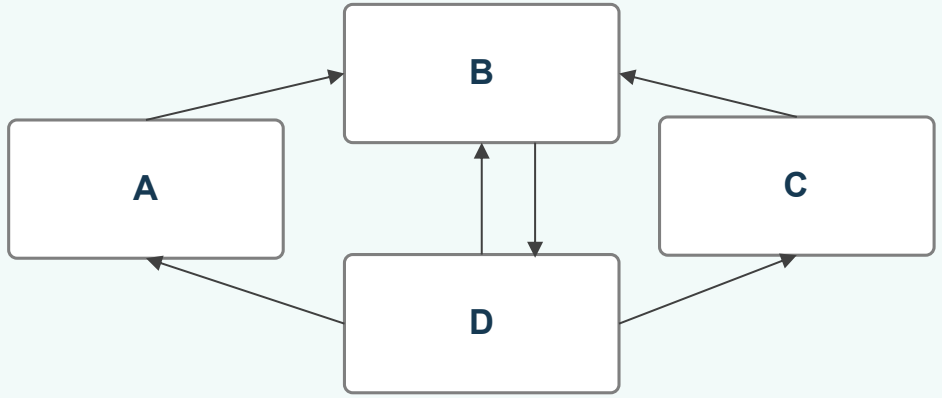
Integrated operating authority



One actor can mandate integration, testing and change.

Distributed/Fragmented ecosystem authority

Company / Operator / Actor



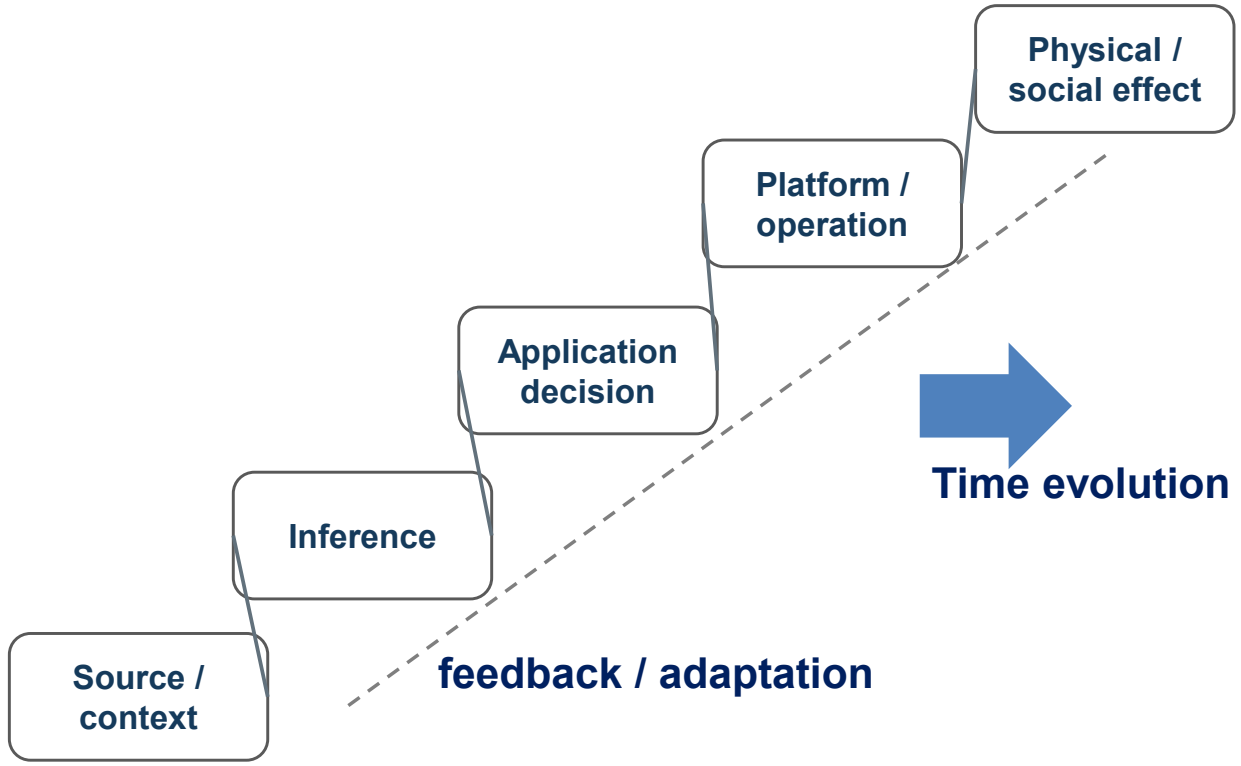
Agreements and delegated rights must make joint action possible.

Collective action becomes governable through agreed rights, shared evidence and executable decisions.

02 Effects propagate across layers and time

Data, inference, decisions, actions and physical effects can become the next system's input.

L5	Physical / Visible Ecosystem & Interactions
L4	Shared Infrastructure, Platforms & Orchestration
L3	Operational Systems & Applications
L2	Decision Logic, Models & Algorithms
L1	Data, Signals & Knowledge Context
L0	Governance & Operating Conditions

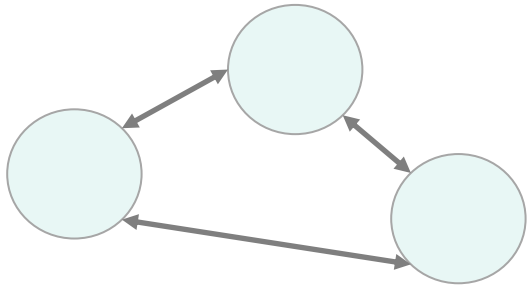


Data and actions propagate between layers;
authority and obligations must also reach the shared effect.

03 Multi-Agent Phenomenon

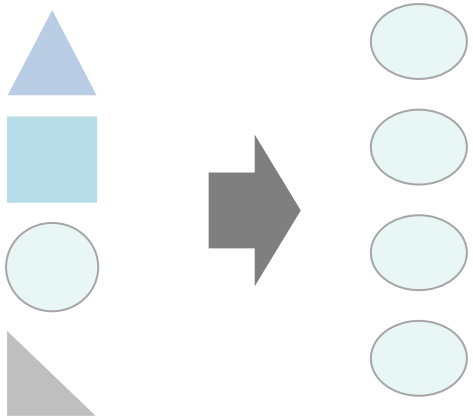
Collective behaviour can emerge even when each local actor is following a reasonable rule.

Coordination lock-in



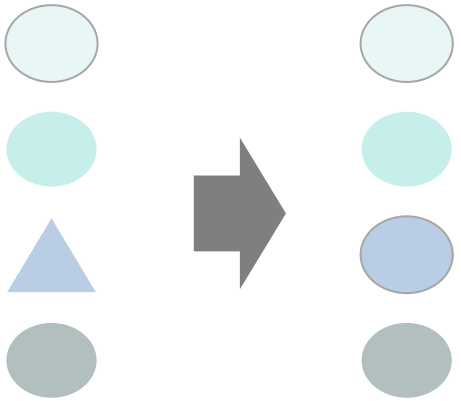
Incentive convergence
Optimisation pressure
Over-compliance

Diversity collapse



Local optimisation

Outlier suppression



Filtering
Engagement optimisation
Policy moderation

Collective outcome may be nobody's local intent.

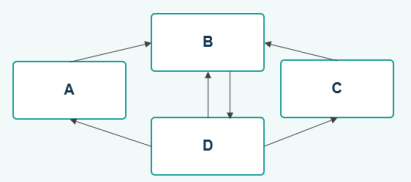
What Makes Ecosystem Governance Different?

Governance should extend beyond individual actors and systems to the ecosystem-level conditions created by their interactions.

Three structural challenges of Ecosystem

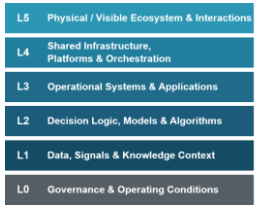
01 · Distributed / Fragmented Authority

Roles, information and decision rights span organisational boundaries.



02 ·Cross-system and time Propagation

A local response can change conditions elsewhere, across actors and layers.



03 · Multi-actor Interaction Phenomena

Collectively undesirable outcomes can emerge from reasonable local decisions.



Healthy actors do not guarantee a healthy ecosystem.

When locally reasonable actions interact

The challenge is not limited to faulty components or rule-breaking.

LOCAL DECISION

A robot stops conservatively.

An AI optimises a route.

A care system reduces workload.

INTERACTION

The responses become other actors' constraints, signals and incentives.

ECOSYSTEM OUTCOME

Standstill, concentrated risk or an emerging concern that is missed.

Healthy actors do not guarantee a healthy ecosystem.

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What is the condition to govern the whole ecosystem

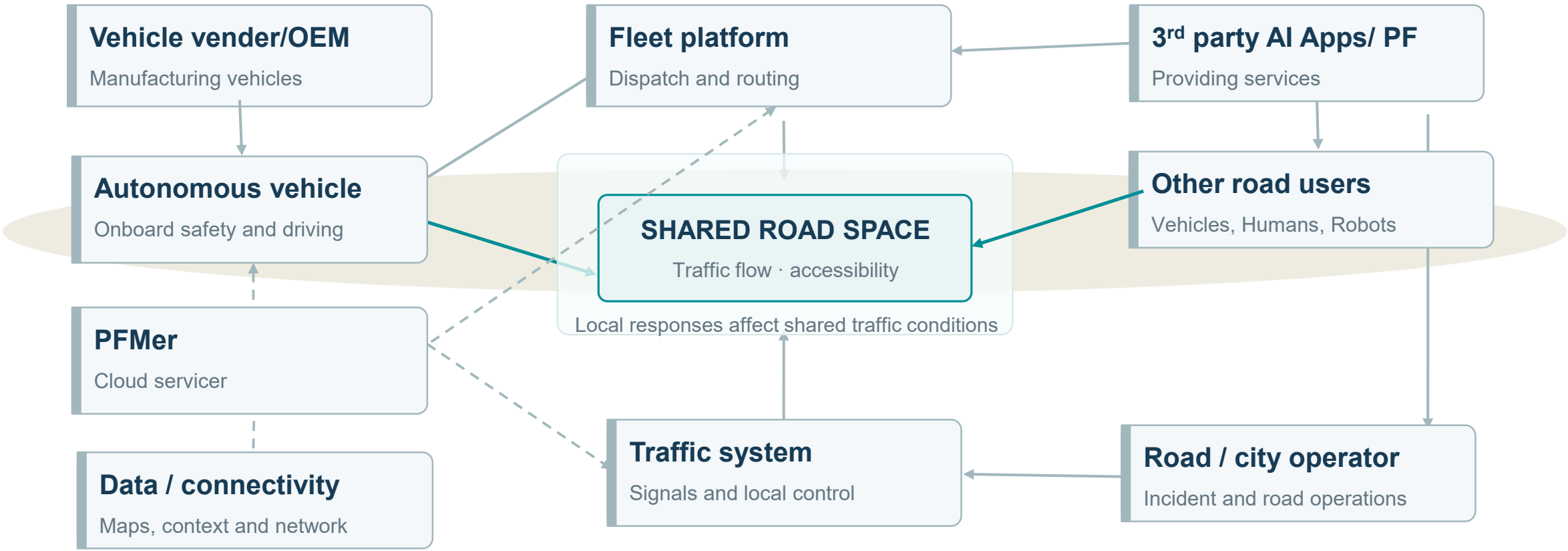
What is needed to govern the ecosystem?

More rules?

More powerful authority?

One Mobility Outcome, Multiple Interdependent Systems

Vehicles, platforms, infrastructure and road users shape a shared mobility outcome.



The shared outcome depends on systems and actors beyond any single vehicle.

Responsibility as a Societal Safety Mechanism

Its historical governance effect was anticipatory: the prospect of a future reckoning disciplined present conduct.

CREDIBLE EXPECTATION	CURRENT DECISION	PREVENTIVE EFFECT
Later scrutiny, attribution and consequences are foreseeable	A party has a reason to moderate its present conduct	The prospect can deter unsafe or harmful action

The sanction does not have to occur often; the credible prospect can already shape behaviour.

Responsibility as an Anticipatory Governance Mechanism

Distribution and Fragmentation Are Structurally Co-produced

Two distinct effects arise from the same ecosystem configuration; neither effect logically entails the other.

SHARED INSTITUTIONAL CONFIGURATION

Control and benefit concentrated • Obligations externalised • Provider/recipient roles unsettled

DISTRIBUTION

Responsibility does not settle on a canonical point of imputation.

Not merely “many responsible actors”

FRAGMENTATION

The relation itself is split: control, causal contribution, capacity, benefit, answerability.

Not merely splitting the outcome into parts

One Institutional Form - Without Presuming Virtue or Capability

Minimum constraints, active maintenance and ecosystem-level noticing are one non-unique arrangement - not fixed substantive rules.

ONE POSSIBLE ARRANGEMENT

Minimum constraints

A binding protective floor across participants.

Positive maintenance

Active upkeep of the shared field; non-transgression alone is insufficient.

Ecosystem-level noticing

Perceive emergence, collective drift and transitions; initiate response.

WHAT IS NOT PRESUMED

Virtue / goodwill

No assumption that participants are well disposed.

Internal sense of obligation

No reliance on actors feeling bound.

Capability

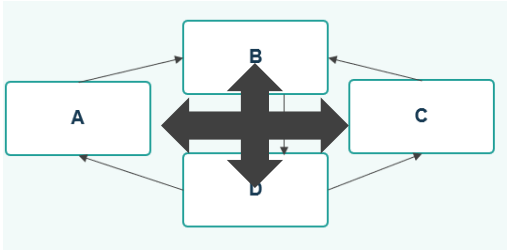
Not presumed as present; required to perform maintenance.

Governance Design

A Governance Thread connects the actors, mechanisms and decisions needed to make ecosystem-level conditions effective as the ecosystem evolves.

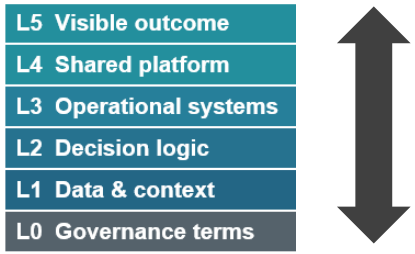
Design Governance Thread

Across Actors



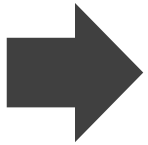
Who is **responsible**?
Whose **decisions, information, approvals and cooperation** are required?

Across layers



Are each layer **connected** where needed?

Across time & Change



Will the conditions remain effective through **updates, interventions, propagation, stops, restarts and changes**?

For each Thread, identify the shared condition, participating actors, decision authority, and mechanisms that make it operational.

What should the ecosystem govern?

Three Requirements for Ecosystem Governability

DEFINE THE FLOOR

What common minimum operating conditions must hold across the ecosystem?

1

Establish **the minimum conditions, shared operating assumptions and boundaries** that must hold across the ecosystem.

SUSTAIN GOVERNABILITY

Who keeps these conditions effective as systems and relationships change?

2

Maintain the capabilities, authority, relationships and mechanisms needed to keep these conditions effective as actors, systems and circumstances change.

ENABLE NOTICING

How does emerging erosion, drift or failure become visible to the relevant actors?

3

Make erosion, drift and emerging failures of ecosystem-level conditions detectable.

Reference:

[Real Case] When Locally Safe Behaviour Creates a System-Level Problem

2025/ San Francisco USA

A fire at a power station cut electricity to about one-third of San Francisco, so the traffic lights went dark. Waymo's self-driving cars were built to treat a dark light like a stop sign. That caused the heavy traffic jams.

Reasonable local rule
Treat dark traffic signals cautiously and request confirmation "Is it OK to go?" when needed.

Large-scale disturbance
Power outage → simultaneous confirmation requests → support backlog

Unintended collective effect
A precaution that works locally can contribute to congestion when many vehicles respond together.

[Use Case] Technology × Governance: Designing for System-Level Disruptions

Illustrative redesign inspired by the 2025 San Francisco outage — not a description of Waymo's actual architecture.

Technology

- Detect widespread infrastructure failures
- Monitor fleet-wide effects and support capacity
- Enable emergency operating modes, pausing and rerouting
- Test recovery under large-scale disruptions



Governance Design

- Define acceptable conditions for shared road use
- Set triggers for exceptional operations
- Establish intervention rights and coordination with city authorities
- Define recovery and restart conditions

A Governable Operating System

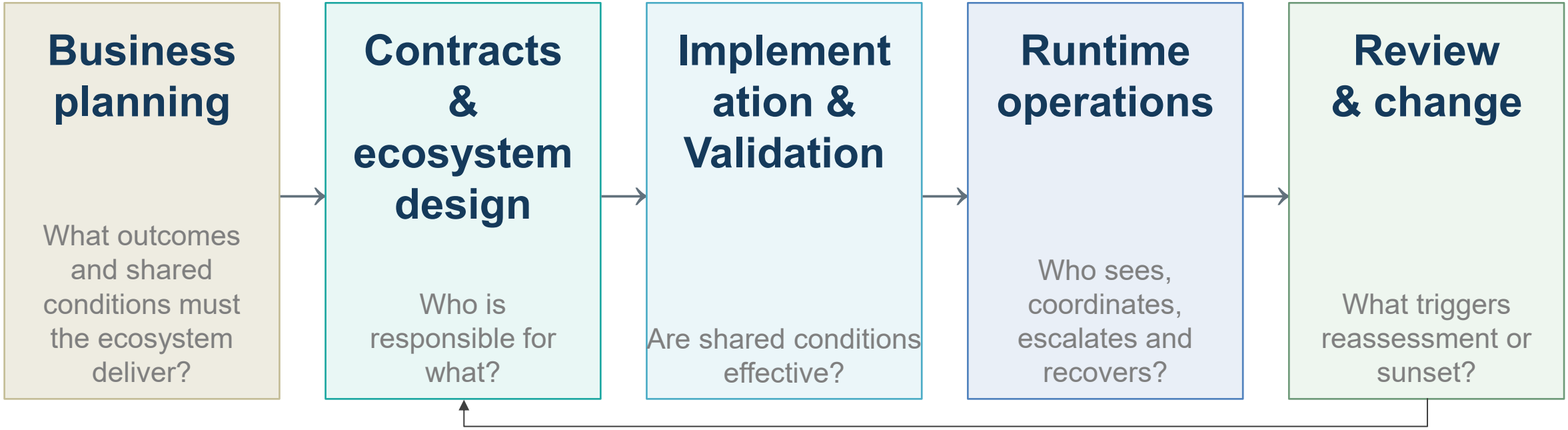
Detect changing conditions, coordinate interventions, and maintain acceptable shared outcomes.

5

Summary & Open research Questions

Governability Must Be Designed from the Start to Runtime

Ecosystem governability cannot be added as an afterthought. It must shape business planning, ecosystem arrangements and implementation — and remain effective through operation, review and change.



Changes in the ecosystem must feed back into its governance design

From Intelligent Agents to Intelligent Interactions

Recent research increasingly addresses what happens when autonomous systems interact.

Logic-Based Agents & Runtime Self-Checking
Reasoning, monitoring and self-checking in evolving agent systems.
Costantini (2022); Vozna et al. (2026)
University of L'Aquila

Multi-Agent Risks
Miscoordination, conflict, collusion and collective risks.
Hammond et al. (2025)

Formal Verification & Contracts
Compositional verification of local objectives and shared safety requirements.
Dewes & Dimitrova (AAAI 2025)

Coordination & Evaluation
Evaluating cooperation, competition and coordination structures.
Zhu et al. (ACL 2025)

Research on individual agents and collective interactions offers complementary foundations for real-world autonomous systems.

Making Autonomous Ecosystems Governable: Technology × Governance

Technical capabilities and governance arrangements must be designed together.

Technology

- **Specification & Verification**
  Logic-based reasoning, formal methods
- **Runtime Monitoring**
  Complex event processing, distributed observability
- **Coordination & Adaptation**
  Multi-agent protocols, adaptive control
- **Intervention & Recovery**
  Fallback, safe restart, re-verification



Governance Design

- **Shared Conditions**
  Objectives, acceptable limits, operating assumptions
- **Evidence & Oversight**
  Information sharing, verification obligations
- **Authority & Coordination**
  Decision rights, roles, cross-organisational commitments
- **Accountability & Change**
  Escalation, restart authority, review and revision

Neither technology nor governance arrangements are sufficient on their own.

Open Research Questions for Governable AI Ecosystems

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01 Representation

How can we specify shared conditions that do not belong to any single agent or system?

Knowledge representation · Formal specification · Temporal logic

02 Composition & Verification

When can local guarantees be combined into system-level assurances — and when do the underlying assumptions cease to hold?

Compositional verification · Assume–Guarantee reasoning

03 Runtime Detection

How can we detect harmful collective behaviour or the erosion of shared conditions when no actor observes the whole ecosystem?

Complex event processing · Distributed monitoring · Anomaly detection

04 Coordinated Intervention & Adaptation

How can independent actors coordinate intervention, recovery and revision without relying on a single controller?

Multi-agent protocols · Decision rights · Human–AI coordination

When Answers Get Cheap, What Becomes Worth Building?

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Three things to take away

1. AI changes what creates value.

Beyond efficiency and automation, AI enables new possibilities for context-specific and adaptive solutions. Framing problems, choosing objectives and exercising judgment become increasingly important.

2. Value and risks emerge from interactions.

The next industrial transformation extends beyond individual products and organisations. Intelligent components alone do not guarantee desirable ecosystem outcomes.

3. Making it work requires Technology × Governance.

Technical capabilities and governance arrangements must evolve together to define shared conditions, detect change, coordinate responses and adapt over time.

The next frontier is exploring what intelligent systems can achieve together — and how we can make that possible in the real world.

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→Paper / Insight

The Institutional Conditions of Governance in Ecosystems

https://dx.doi.org/10.2139/ssrn.7242779

From Localized Causality to Propagation Dynamics: Rethinking AI Governance Assumptions

http://dx.doi.org/10.2139/ssrn.6776720